CyberTwin AI Dashboard Project Documentation

Project Name: CyberTwin AI Dashboard
Technology Stack: Python, Streamlit, Machine Learning, Plotly, IMAP Email Integration
Project Type: AI Powered Cybersecurity Simulation Platform

1. Introduction

CyberTwin AI Dashboard is an advanced cybersecurity simulation platform developed using Python and Streamlit. The project focuses on detecting cybersecurity risks, analyzing password strength, detecting spam emails, and simulating risk analysis using Artificial Intelligence and Machine Learning techniques.

The dashboard provides an interactive and modern user interface where users can connect multiple online platforms, evaluate cyber risks, analyze emails, and visualize security metrics in real time.

2. Objectives of the Project

- To create a cybersecurity awareness platform.
- To simulate cyber risk analysis using AI models.
- To detect spam and phishing-related emails.
- To analyze password strength and security.
- To provide an attractive and interactive dashboard UI.
- To visualize cyber threats using charts and analytics.

3. Technologies Used

Technology	Purpose
Python	Core Programming Language
Streamlit	Web Dashboard Development
Pandas	Data Handling
NumPy	Numerical Computation
Scikit-learn	Machine Learning Models
Plotly	Interactive Graphs and Charts
IMAPLib	Fetching Emails from Gmail
NetworkX	Graph Visualization

4. Major Features

- AI-based cyber risk prediction
- Password security checker
- Spam email detection system
- Email fetching using Gmail App Password
- Multi-platform account analysis
- Interactive charts and metrics
- Glassmorphism modern UI design
- CSV export functionality

5. Project Workflow

1. User opens the Streamlit dashboard.
2. User connects different platforms like Email, Facebook, Instagram, etc.
3. The system generates synthetic cybersecurity data.
4. Machine learning models analyze the risk level.
5. Password strength is evaluated.
6. Emails are fetched securely from Gmail using IMAP.
7. Spam detection logic classifies emails as Spam or Safe.
8. Analytics and graphs are displayed on the dashboard.

6. Important Code Snippets

6.1 Streamlit Page Configuration

```
st.set_page_config(
    page_title="CyberTwin AI Dashboard",
    page_icon="🛡",
    layout="wide"
)
```

6.2 Password Strength Checker Logic

```
if len(password) >= 10:
    score += 1

if any(c.isdigit() for c in password):
    score += 1

if any(c.isupper() for c in password):
    score += 1
```

6.3 Spam Detection Function

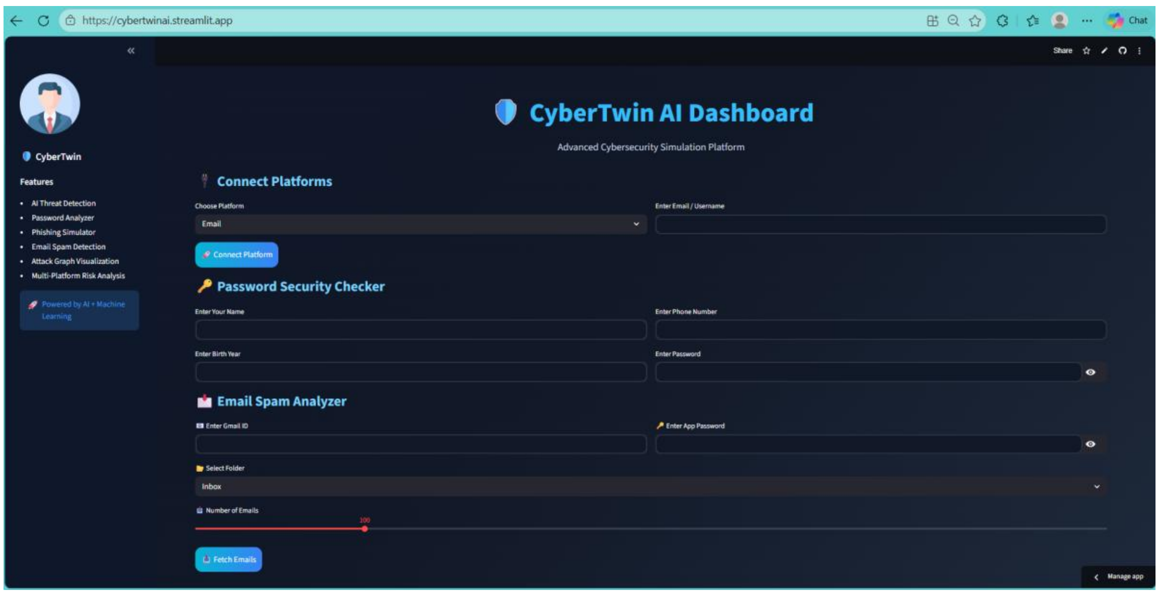
```
def detect_spam(text):

    spam_keywords = [
        "free", "win", "urgent",
        "click", "offer", "money"
    ]

    score = sum(
        word in text
        for word in spam_keywords
    )

    return "Spam" if score >= 2 else "Safe"
```

7. Machine Learning Models Used

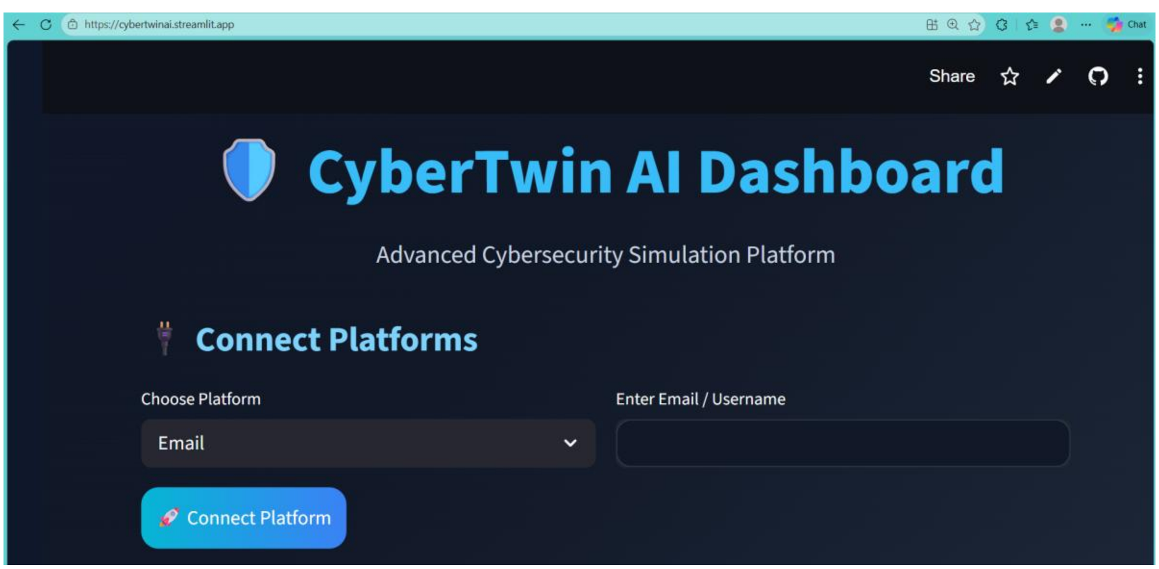
- Logistic Regression
- Random Forest
- Decision Tree
- Gradient Boosting
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Extra Trees Classifier
- AdaBoost Classifier
- Gaussian Naive Bayes

8. Screenshots Section

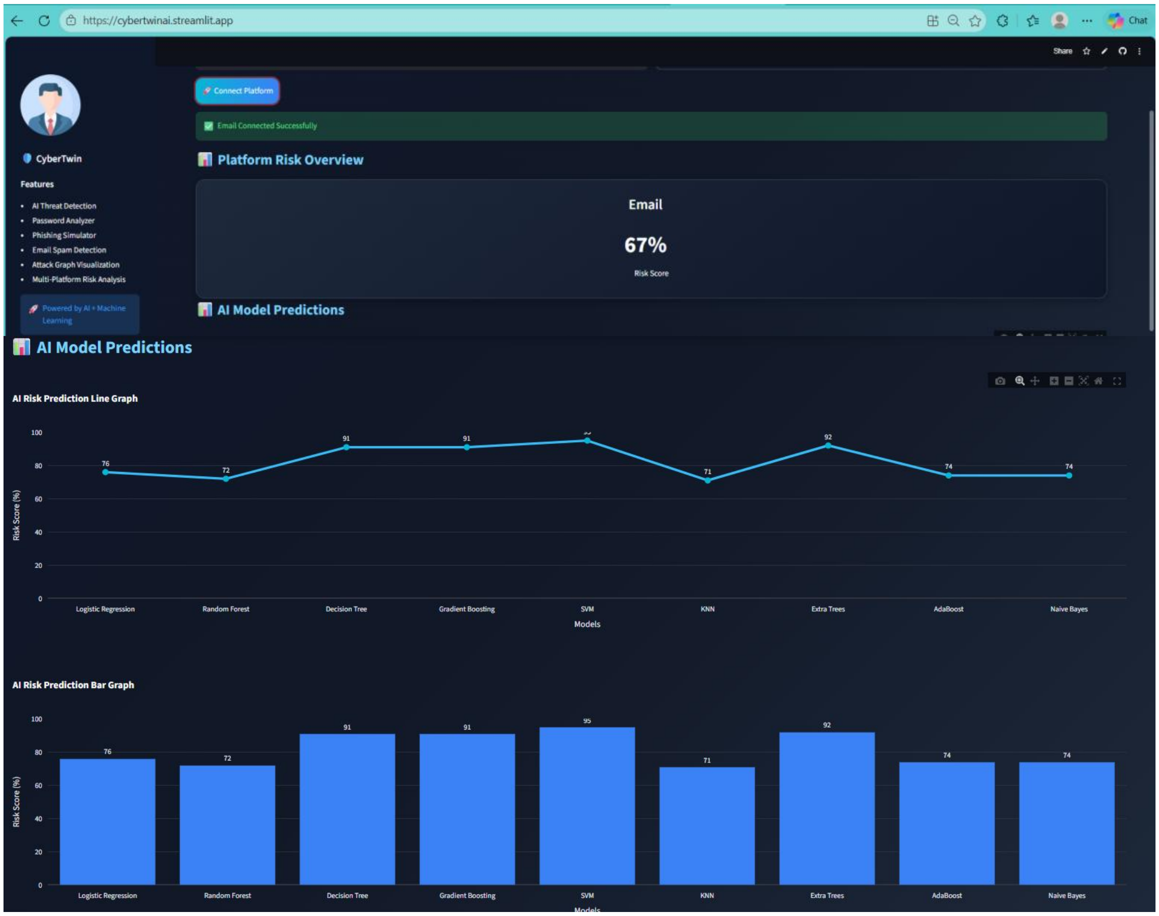
Homepage Dashboard Screenshot



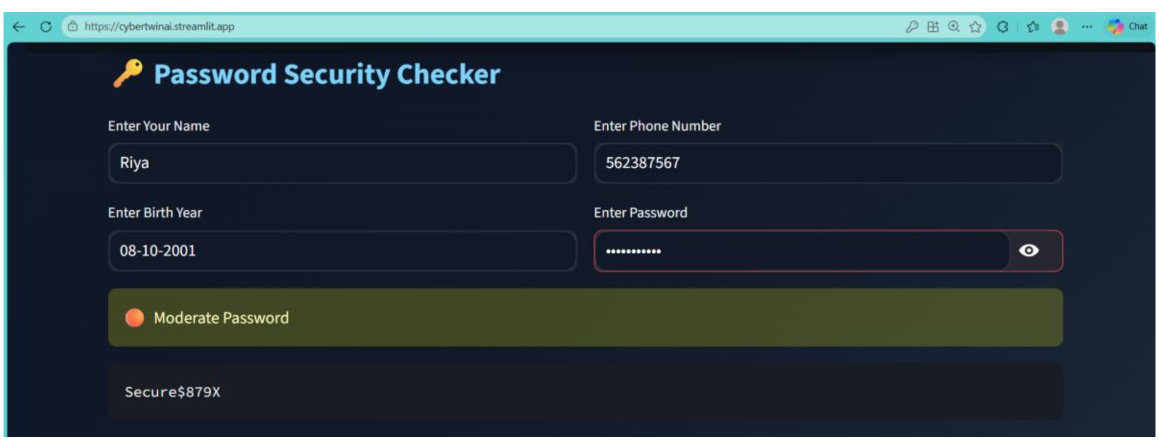
Platform Connection Screenshot



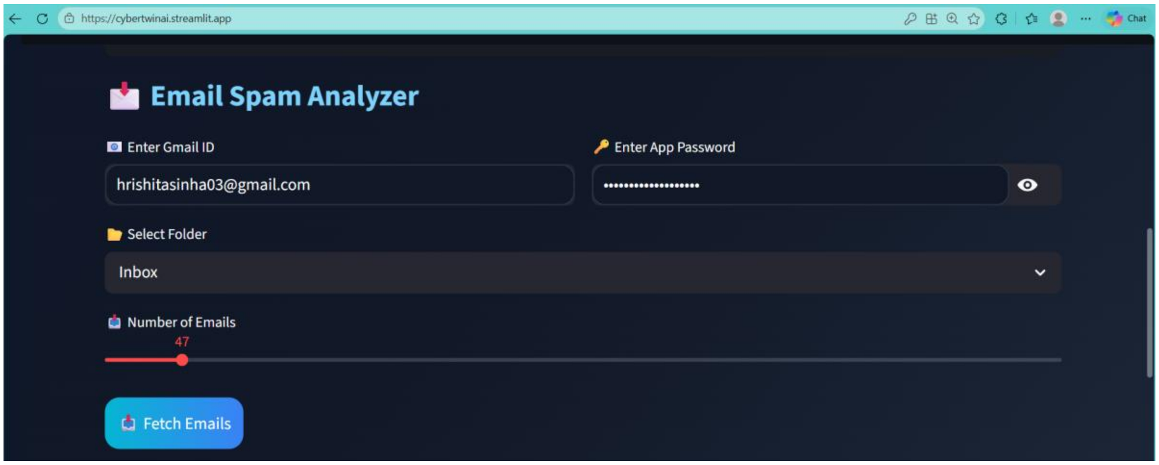
Risk Analysis Graph Screenshot

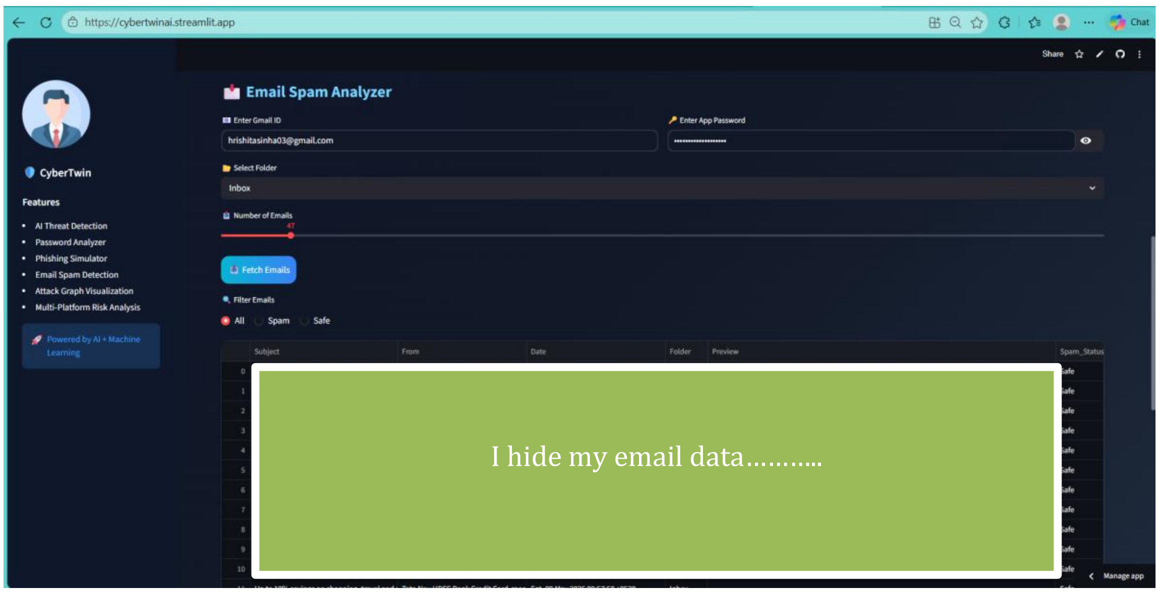


Password Security Checker Screenshot

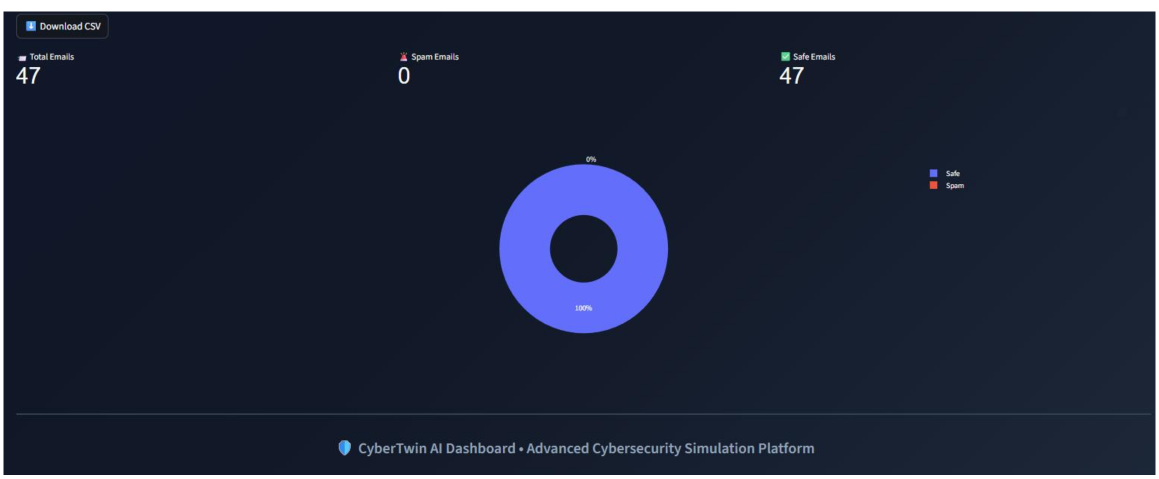


Email Spam Analyzer Screenshot





Spam vs Safe Pie Chart Screenshot



9. Challenges Faced

- Handling Gmail IMAP authentication
- Fetching emails from different folders
- Designing modern responsive UI
- Managing Streamlit session state
- Implementing spam detection efficiently
- Training multiple ML models

10. Future Enhancements

- Real-time phishing detection using NLP
- Dark Web leak detection
- Live threat monitoring system
- Cloud database integration
- User authentication system
- Deployment using Docker and Cloud Services

11. Conclusion

CyberTwin AI Dashboard is a complete AI-powered cybersecurity simulation platform that combines machine learning, email analysis, password security, and interactive visualization into one modern application. The project demonstrates practical implementation of cybersecurity concepts using Python and modern web technologies.

12. Acknowledgement

I would like to express my sincere gratitude to my mentor for providing valuable guidance, continuous support, and encouragement throughout the development of the CyberTwin AI Dashboard project. Their suggestions and technical insights helped me improve the quality of the project and gain a deeper understanding of cybersecurity, machine learning, and dashboard development.

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